Ryan Peckham

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Education

University of Michigan — Ann Arbor, MI

B.S. Computer Engineering and Applied Physics; GPA: 3.35/4.0 — Aug. 2025 – Apr. 2029

Coursework: Multivariable & Vector Calculus, Electricity & Magnetism, Quantum Photonics, Discrete Mathematics, Solar Energy Systems, Mandarin, Probabilistic Statistics

Skills

Programming Languages: Python, C, C++, C#, Rust, Java, SQL, Front-End Stack, MATLAB
Systems & Concepts: Git & GitHub, PostgreSQL, WSL, Docker, Database Normalization, Threading
Operating Systems: Windows, Linux (Ubuntu, Debian, NixOS), MacOS
Hardware: Soldering, Shielded Metal Arc Welding, CNC Machines, Lathes, Mills

Experience

Michigan Autonomous Security Team (M-AST), Chief Software Engineer — Jan. 2026 – Present

University of Michigan

- Leading and instructing a team of engineers to install, program, and test gyroscope, magnetometer, barometer, and GPS sensors onto a prototype drone
- Maintaining the organization of a large and complex codebase and reviewing code and pull requests
- Collaborated with M-AST President and peers to design the drone architecture specifying the flow of information between systems, including the implementation of SITL and HITL runtimes
- Integrated several software architectures including ROS2, OpenCV, telemetry code, and path-finding

Michigan Hackers, Machine Learning Team, Member — Nov. 2025 – Present

University of Michigan

- Learned and applied fundamental computer science concepts related to machine learning such as Markov Chains, Bayes' Theorem, and the various kinds and applications of machine learning
- Attended regular lectures by other students while discussing and practicing new material with peers

Engineering 100 — Aug. 2025 – Dec. 2025

University of Michigan

- As part of a 5-member engineering team, completed the C++ programming, wiring, and soldering of a sun-tracking solar panel using photoresistors, stepper motors, voltage limiters, and a battery on a Raspberry Pi platform
- Engaged in a comprehensive engineering process including identifying a problem in modern technology, thoroughly researching the subject, and presenting a proposed solution to Engineering faculty; tested the solution and presented results, data, and conclusions to faculty
- Developed skills in communication, collaboration, task prioritization, concluding meaningful results from data, and balancing planning with execution

FIRST Robotics Competition (FRC), Team 141 "WOBOT" — Aug. 2023 – Oct. 2024

West Ottawa High School, Programming Sub-team

- Developed input and movement code by implementing and tuning PIDF loops for precise control
- District Champions; awards won by team: Engineering Inspiration, Sustainability, Entrepreneurship

Cultural, Technological, and Environmental Exchange (CTEE) Club — Sep. 2024 – Apr. 2025

West Ottawa High School

- Repaired and tested 2–3 community-donated desktop computers and monitors each week
- Learned about the architecture and design of desktop computers & motherboards
- Replaced faulty components safely and prepared ready-to-use operating systems (Windows)
- Stored repaired units for future shipment to developing nations and communities

Personal Projects

- Designed and developed my personal website peckhamr.com using HTML, JS, CSS, and the Svelte UI framework, as an opportunity to teach myself the foundations of domain registrars, website hosting, and maintainable codebase design
- Applied the skills I learned to create the public website for M-AST, flymast.com; a more professional and sleek design built for prospective students and sponsors